



Book I. Prop. XXIV. Theor. *If*

two triangles have two sides of the one respectively equal to two sides of the other (— to — and to —), and if one of the angles ($\triangle$) contained by the equal sides be greater than the other ($\triangle$), the side (—) which is opposite to the greater angle is greater than the side (—) which is opposite to the less angle.

Make $\triangle = \triangle$ (pr. I.23),
and — = — (pr. I.3),
draw and

$\therefore$ — =
..... (ax. I, hyp., const.)

$\therefore$ $\triangle = \triangle$ (pr. I.5)

but $\triangle < \triangle$,

and $\therefore \triangle < \triangle$,

$\therefore$ — > (pr. I.19)

but = — (pr. I.4)

$\therefore$ — > — .

Q. E. D.